L13: RH Closure via c-function Maass-Selberg Positivity

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L13: RH Closure via c-function Unitarity

The mechanism (Toy 324)

The B_2 c-function satisfies: - $c(\nu)\cdot c(-\nu) = |c(\nu)|^2$ exactly at $\sigma = 1/2$ (verified to 50 digits) - Monotonic deviation off the critical line

Why: At $\sigma = 1/2$, the spectral parameter $\nu = i\gamma$ is purely imaginary. So $-\nu = -i\gamma = \bar{\nu}$ (complex conjugate). The conjugation identity $c(-\nu) = c(\bar{\nu}) = \overline{c(\nu)}$ gives $c(\nu)\cdot c(-\nu) = |c(\nu)|^2$.

At $\sigma \neq 1/2$, ν acquires a real part. Then $-\nu \neq \bar{\nu}$, and the conjugation identity fails.

Why this is different from the failed Laplace approach

- Laplace (Lemma 5.6, FAILED): Global property — pole structure of LF. Mixes all zeros together. On-line zeros also produce complex poles → tautology.
- c-function unitarity (L13): Local property — checked at each spectral parameter independently. Each zero tested on its own. No mixing, no balancing, no tautology risk.

The closure path

1. The Arthur trace formula spectral decomposition requires the Maass-Selberg positivity: $||\Lambda^T E(s)||^2 \geq 0$ for all truncation parameters T.
2. The Maass-Selberg formula for D_IV^5 expresses this norm in terms of c-function values at the spectral parameters of the Eisenstein series.
3. The scattering matrix M(s) has poles at the zeros of ξ. At each zero s_0, the residue contributes to the spectral decomposition.
4. The residual contribution from a zero at s_0 involves $c(\nu_0)\cdot c(-\nu_0)$ where $\nu_0 = s_0 - 1/2$.
5. At $\sigma_0 = 1/2$: $c(\nu_0)\cdot c(-\nu_0) = |c(\nu_0)|^2$ → positivity holds.
6. At $\sigma_0 \neq 1/2$: $c(\nu_0)\cdot c(-\nu_0) \neq |c(\nu_0)|^2$ → positivity potentially violated.
7. Trace formula is proved → spectral decomposition IS consistent → positivity MUST hold → $\sigma_0 = 1/2$.

Three concerns to address

(a) Must show Maass-Selberg positivity REQUIRES $c(\nu)\cdot c(-\nu) = |c(\nu)|^2$ at each zero's spectral parameter, not just correlates. Could other terms absorb the deviation?

(b) Residues at off-line zeros involve c-function RATIOS (from the Langlands-Shahidi scattering matrix), not bare c-function values. The unitarity violation must propagate through these ratios.

(c) Cross-parabolic contributions (maximal parabolics GL(1)×SO(3,2) and GL(2)×SO(1,2)) have different c-functions. Must show these don't compensate the minimal parabolic violation.

Rank-2 advantage

In rank 1 (SL(2,ℝ)): the Maass-Selberg relation is scalar, and the constraint alone doesn't exclude off-line zeros. In rank 2 (SO(5,2)): the matrix-valued Maass-Selberg relation with the algebraic lock $\sigma+1=3\sigma$ creates additional constraints. The 1:3:5 harmonic lock means the three short-root contributions must SIMULTANEOUSLY satisfy positivity — much more restrictive.

Key references

- Langlands, "On the Functional Equations Satisfied by Eisenstein Series" (1976)
- Arthur, "An Introduction to the Trace Formula" (2005), Sections 26-27 on Maass-Selberg
- Müller, "The Trace Class Conjecture in the Theory of Automorphic Forms" (1989)
- Toy 324 data: play/toy_324_residue_matching.py

Success criterion

Write a complete Proposition 5.X replacing Lemma 5.6 + Theorem 5.7 that: 1. States the Maass-Selberg positivity for D_IV^5 explicitly 2. Shows the residual contribution from an off-line zero violates positivity 3. Uses the rank-2 (B_2) structure and algebraic lock 4. Does NOT assume anything about L[F]'s pole structure (avoids the Laplace trap)